Topic 2: Physical Layer

Digital Transmission in Computer Networks

- The purpose is to transfer a data sequence of 0s and 1s from a transmitter to a receiver
- It uses pulses or sinusoids to transmit binary information over a physical transmission medium

pre-defined signal formats

- discrete symbols
- finite symbol set e.g. {0,1}
- finite # signal formats (pulses)

Digital Transmission in Computer Networks

- We are particularly interested in the bit rate measured in bits/second

bps

$$\uparrow \text{bit rate} = \frac{\#\ \text{bits}}{\text{second}} = \frac{\#\ \text{bits}}{\text{signal format (pulse)}} * \frac{\#\ \text{pulses}}{\text{second}}$$

(channel) Coding rate
- # pulses
- pulse shape

Baud rate
- transmission medium
- channel bandwidth (BW)
- pulse shape

Bit Rate vs. Baud Rate

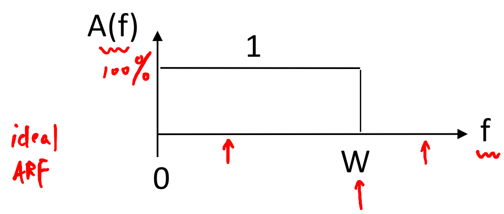
- Definitions
  - Bit Rate = # of bits transmitted per second
  - Baud Rate = # of signal transitions per second
- Baud Rate depends on the channel bandwidth
- Bit Rate = (Baud Rate) × (# bits per pulse)
  - It depends on the channel bandwidth as well as the coding scheme

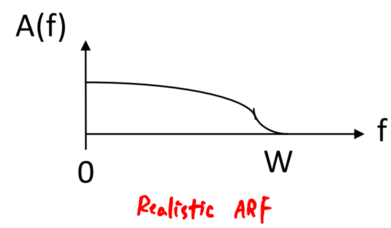
Transmission Channel and Channel Bandwidth

- A transmission channel can be characterized by its effect on input sinusoidal signals (tones) of various frequencies

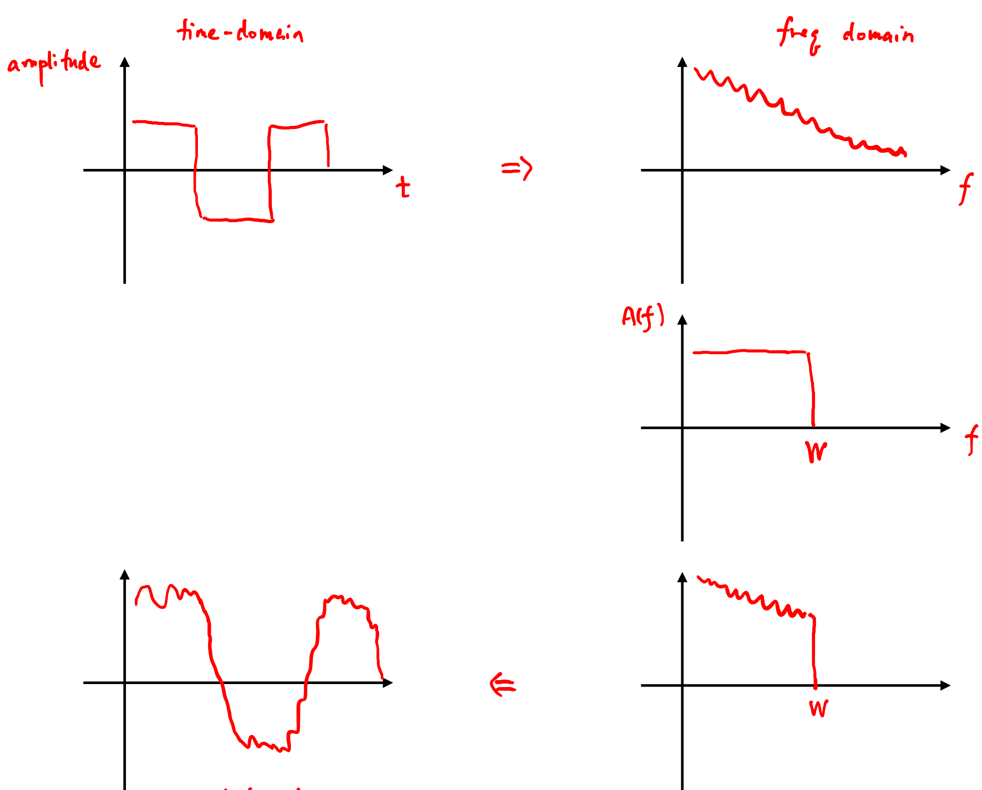
- The ability of the channel to transfer a tone of frequency f is given by the amplitude-response function (A(f)), which is defined as the ratio of the amplitude of the output tone to the amplitude of the input tone

ARF





- The bandwidth of a transmission channel (W) is the range of frequencies that is passed by the channel



- distorted version
- neighbor symbols affecting each other (ISI) : inter-symbol interference

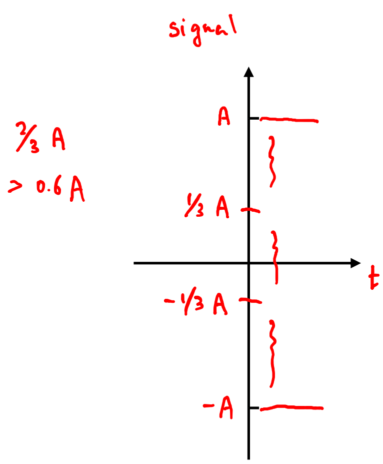
Nyquist Rate

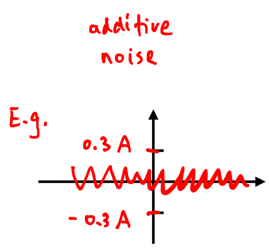
Nyquist Pulses : sinc function : $\text{sinc}(x) = \begin{cases} 1 & x = 0 \\ \frac{\sin x}{x} & \text{else} \end{cases}$

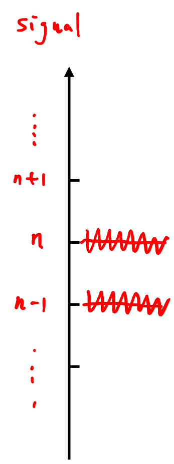
- The fastest rate at which (ideal) pulses can be transmitted over the channel (called the Nyquist Rate) is: max Baud rate (upper bound)

$$r_{max} = 2W \text{ pulses/second}$$

Channel bandwidth







min separation between signal levels $= 0.3A - (-0.3A)$

$= 0.6\,A$

$$\left\lfloor \frac{A-(-A)}{0.6\,A} \right\rfloor + 1 = 4 \text{ levels}$$

$\Downarrow$

2 bits/pulse

Multilevel Pulse Transmission

- Assume channel bandwidth of W
- If pulse amplitudes are either -A or +A, then each pulse conveys 1 bit,
  Bit Rate = (2W pulses/sec) × (1 bit/pulse) = 2W bps
- If amplitudes are from {-A, -A/3, +A/3, +A}, then each pulse conveys 2 bits,
  Bit Rate = (2W pulses/sec) × (2 bits/pulse) = 4W bps
- By going with $M = 2^m$ amplitude levels, we achieve
  Bit Rate = (2W pulses/sec) × (m bits/pulse) = 2mW bps
- In the absence of noise, the bit rate can be increased without limit by increasing the pulse level m

Noise & Reliable Communication

- All physical systems have noise
  - Electrons always vibrate at non-zero temperature
  - Motion of electrons induces noise
- Presence of noise limits the accuracy of measurement of received signal amplitude
- Noise places a limit on how many amplitude levels can be used in multilevel pulse transmission
- Errors occur if signal separation is comparable to noise level
- Bit Error Rate (BER) increases with decreasing Signal-to-Noise Ratio (SNR)

SNR ↑ ⇒ good channel ⇒ BER ↓

↓ bad ↑

SNR = $\dfrac{\text{avg signal power}}{\text{avg noise power}}$

(dB)

SNR = 10 dB = $10 \log_{10}$ SNR

⇒ SNR = $10^{10/10}$

= 10

E.g. SNR = 10000

⇒ SNR (dB) = $10 \log_{10} 10000$

= 40 dB

Shannon Channel Capacity

max bit rate
in AWGN channel
(Additive White Gaussian Noise)

$$C = W \log_2 (1 + \text{SNR}) \text{ bps}$$

(Hz) absolute value

- Channel Bandwidth (W) & Signal to Noise Ratio (SNR) determine C

- If transmission rate R > C, reliable communication is not possible
- If transmission rate R ≤ C, arbitrarily reliable communication is possible
  - "Arbitrarily reliable" means the BER can be made arbitrarily small through sufficiently complex coding

- The relation between R_{max} and C is used as a measure of how well a communication system is designed

Example

- Find the Shannon channel capacity for a telephone channel with W = 3.4 KHz and SNR (dB) = 40 dB

$$\Downarrow$$

$$40 \text{ dB} = 10 \log_{10} \text{SNR}$$

$$\Downarrow$$

$$\text{SNR} = 10^4 = 10000$$

$$C = W \log_2 (1 + \text{SNR})$$

$$= 3.4 \text{ kHz} \cdot \log_2 (1 + 10000)$$

$$= 45.2 \text{ kbps}$$

Example

W

- Consider an AWGN channel with a bandwidth of 27 KHz and SNR = 35 dB. Is it possible to transmit reliably over this channel at R = 350 Kbps?

$SNR\,(dB) = 35\ dB = 10 \log_{10} SNR$

$\Rightarrow SNR = 10^{35/10} = 3162$

$C = W \cdot \log_2 (1 + SNR)$

$= 27\ KHz \cdot \log_2 (1 + 3162)$

$= 313.93\ kbps$

$\wedge$

$R = 350\ kbps$

So: impossible

$R = 310\ Kbps$?

possible

Example

W

- Consider an AWGN channel with a bandwidth of 27 KHz. To transmit reliably over this channel at a rate of 200 Kbps, what is the minimum required SNR (in dB)?

R

$C \geq R$

$W \cdot \log_2 (1 + SNR) \geq R$